Simulating Dice Rolls

a

```
set.seed(45)

rolls <- sample(1:6, 6000, replace = TRUE)

table(rolls)
```

```
rolls
   1    2    3    4    5    6 
1032  993 1034 1010  986  945 
```

###b

```
# expectation is the average or mean in statistics

# E(x) = total number of rolls/total number of die face

# E(x) = 6000/6 = 1000

# The expected value is 1000 which is close to what we
# have on the frequency of the 6000 simulated dice rolls
```

c

```
# x > 1030
# n = 6000
# p(1) = 1/6

p_more_1030 <- pbinom(1030,
                      size = 6000,
                      prob = 1/6,
                      lower.tail = FALSE)
p_more_1030
```

```
[1] 0.1454537
```

```
set.seed(45)

sim <- replicate(10000,sum(sample(1:6, 6000, replace = TRUE) == 1) > 1030)

sum(sim)/10000
```

```
[1] 0.1487
```

```
set.seed(1)
sum(sample(1:6, 6000, replace = TRUE) == 1)
```

```
[1] 981
```

```
set.seed(2)
sum(sample(1:6, 6000, replace = TRUE) == 1)
```

```
[1] 959
```

Wood frog larval abundance in vernal pools

```
p = 0.3
n = 10
```

a

```
dbinom(6, size = 10, prob = 0.3)
```

```
[1] 0.03675691
```

b

```
pbinom(2, size = 10, prob = 0.3)
```

```
[1] 0.3827828
```

c

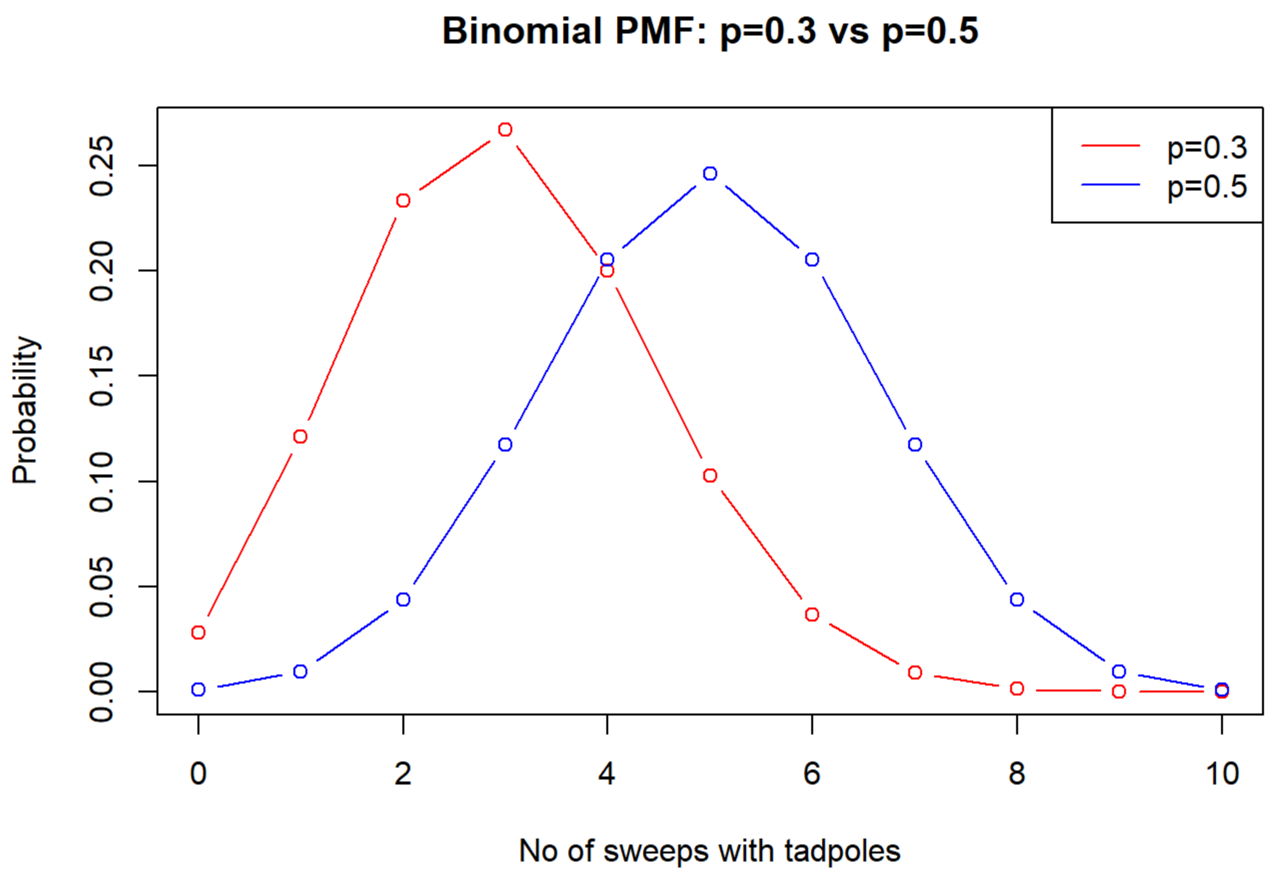
```
pbinom(4, size = 10, lower.tail = FALSE, prob = 0.3)
```

```
[1] 0.1502683
```

d

```
x <- 0:10

plot(x, dbinom(x, size = 10, prob = 0.3),
     type = "b", col = "red",
     xlab = "No of sweeps with tadpoles",
     ylab = "Probability",
     main = "Binomial PMF: p=0.3 vs p=0.5")

lines(x, dbinom(x, size = 10, prob = 0.5),
```

```
      type = "b", col = "blue")

legend("topright", legend = c("p=0.3","p=0.5"),
       col = c("red","blue"), lty = 1)
```



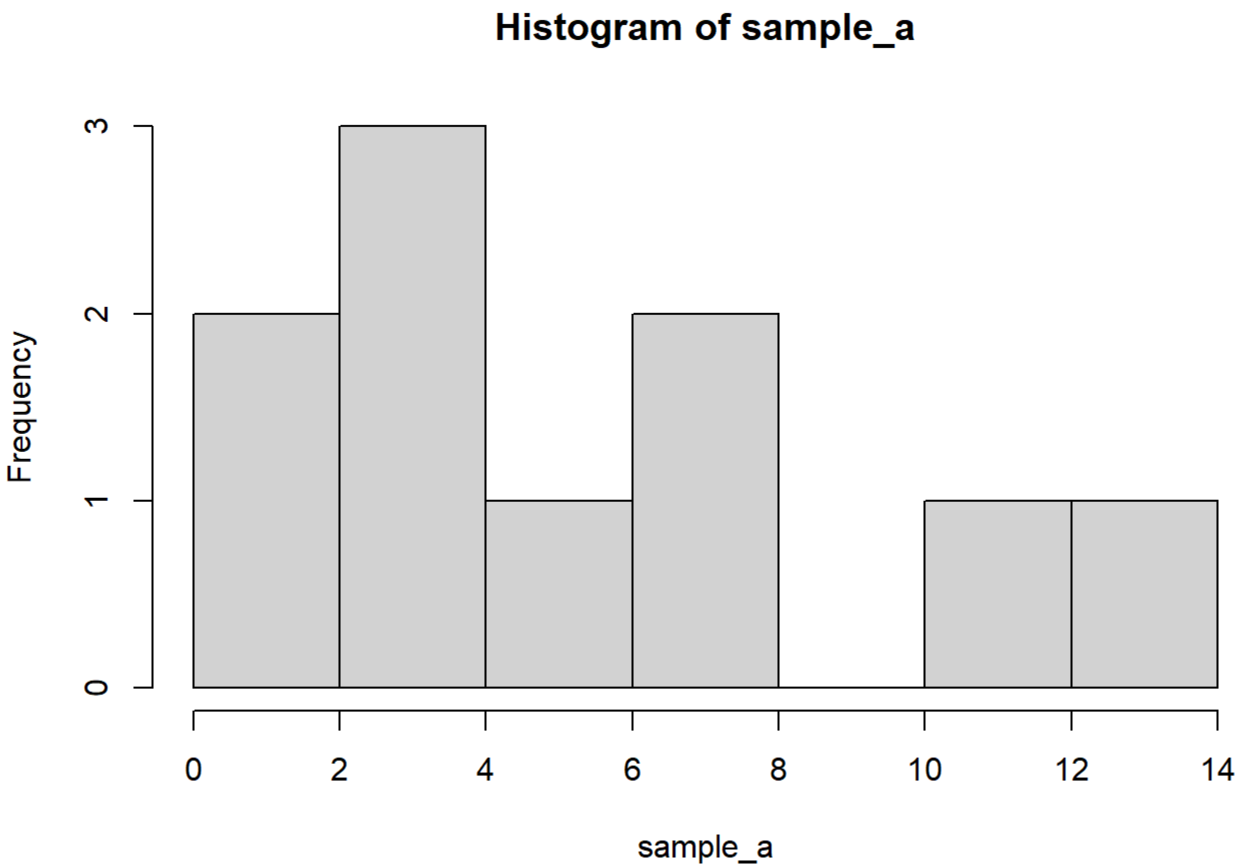
Gamma Distribution

a

```
sample_a <- rgamma(10, shape = 3, scale = 2)

sample_a
```

```
[1]  6.472957  2.628618  1.973382  4.141607  1.045183  7.831022  3.711707
[8] 10.203802  3.744015 13.190905
```

```
hist(sample_a)
```

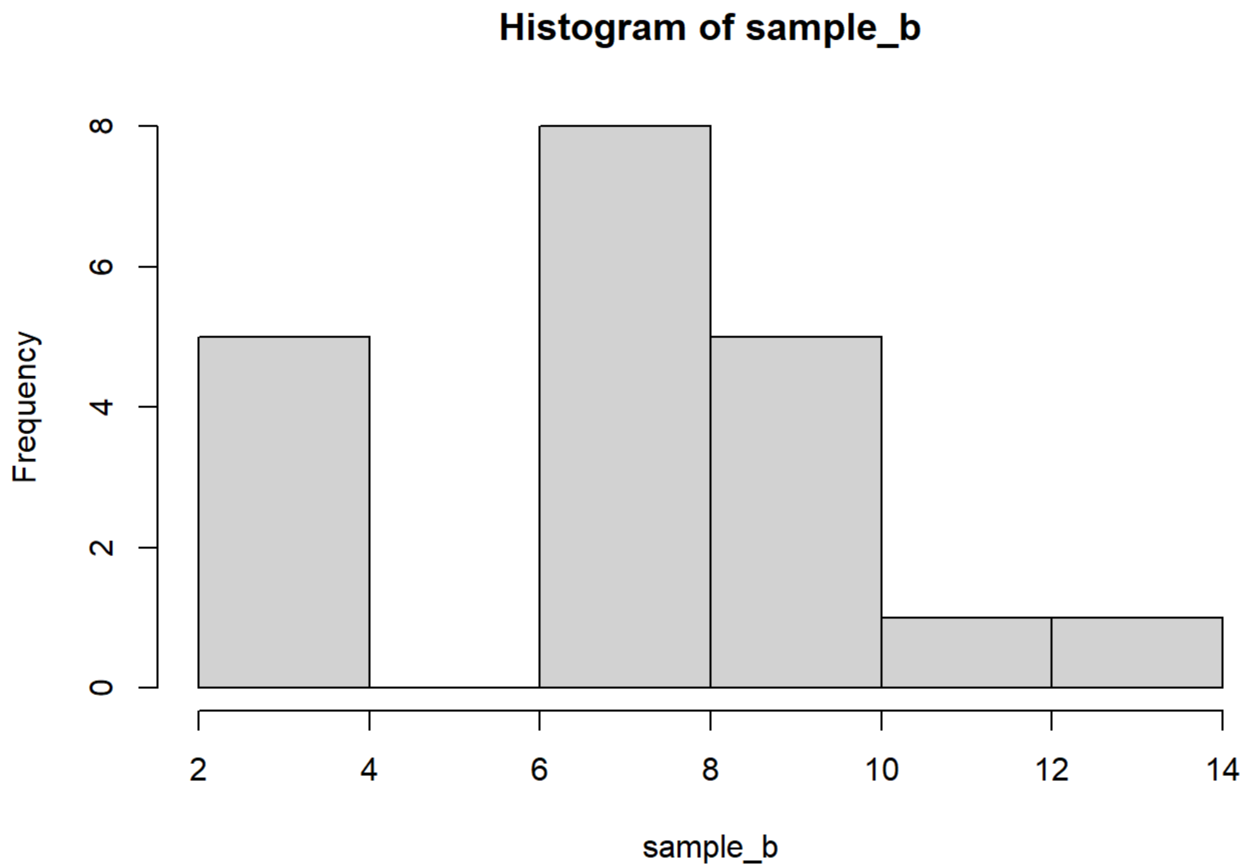


b

```
sample_b <- rgamma(20, shape = 3, scale = 2)

sample_b
```

```
 [1]  8.642270  8.214372  3.832036  2.694994  7.583157  7.179117  6.537613
 [8]  7.205557  6.168363 10.254773  6.925446  8.273957  2.793240  8.759041
[15]  7.790924  3.802486  9.946344  3.075775 12.456139  7.618658
```

```
hist(sample_b)
```



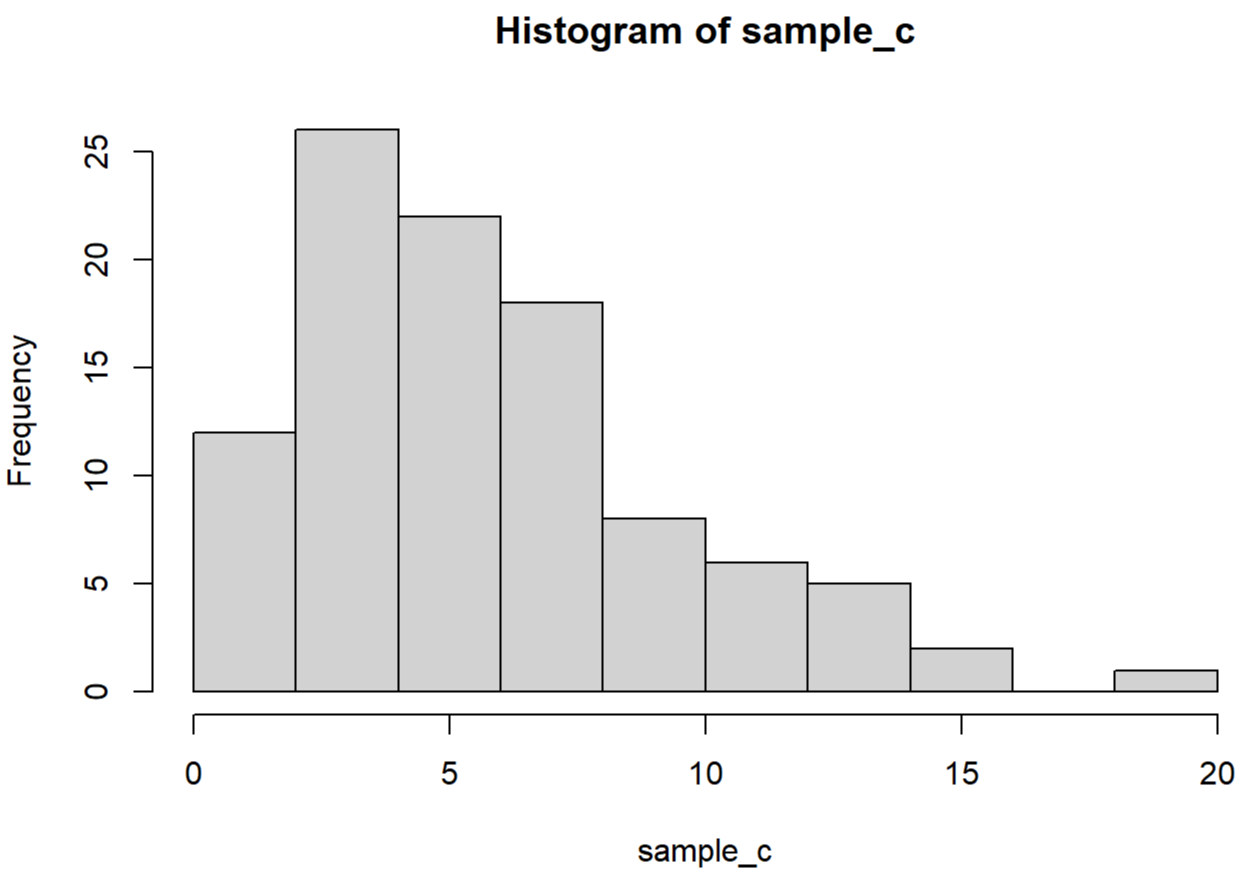
c

```
sample_c <- rgamma(100, shape = 3, scale = 2)

sample_c
```

```
  [1]  4.4470087  6.3392290  3.1171273  0.3087052 18.3050411 10.2892987
  [7] 11.1817437  4.1045669  8.5258405  7.0122387  3.0790206  1.2247356
 [13]  4.0786312  2.3247325  2.7589327  4.9023000  2.5503265  4.2556453
 [19]  2.1438337  9.0822973  0.5600965 11.7942382  6.8126694  7.2568141
 [25] 12.0166676  4.7477227  2.3297922 12.8830416  4.8689574  5.5360207
 [31] 12.4170558 15.6213092  4.1344928  1.2725662  2.1687629  2.1741921
 [37]  8.8216937  3.5314652  6.3768733  9.2742930  1.8072745  2.8880901
 [43]  7.4543134  7.2526139  6.0290045 12.8270201  5.6964932  4.0912485
 [49]  3.6541376  3.3647502  5.3608609  8.9441885  1.0547473  4.7707886
 [55] 10.7439160  3.6615523  5.3023610  6.1095792  1.0940612  1.5501745
 [61]  6.2390839 14.6869256  7.7934673  2.9162660  2.4197987  7.4340084
 [67] 10.9966193  3.1812593  7.1888286  6.5685249  2.1342332 11.4317810
 [73]  6.9449386  4.5008250  4.6192792  3.9790211  6.7705094  0.8624818
 [79]  5.0138163  3.1456983  9.0290268  5.1983588  9.8061019  3.4307839
 [85]  5.0723706  5.3998506  0.6551284  9.0635315  5.0691620  2.6709256
```

```
[91]  1.2888704  4.5057718  2.5814784  2.3896203  2.0881495  7.5678331
[97]  1.0624354  7.8647502  2.4459082 12.8890453
```

```
hist(sample_c)
```



d

```
sample_d <- rgamma(1000, shape = 3, scale = 2)

sample_d
```

```
  [1]  4.9277416  5.1899835  1.9021020  2.9224442  7.8153587 10.6581135
  [7]  1.7798407  3.9063372  7.4684159 16.1103011  4.0135910  2.8230410
 [13] 10.1263993  4.7702957  1.5639549  4.8502189  6.8344817  6.2164402
 [19]  3.6286434 21.1121610  9.9836104  5.0780501  1.8249304  3.9309731
 [25]  6.1112981 11.3099764  2.9379485  8.9831193  2.5073341  4.6724582
 [31]  5.2725918  6.0743426  2.4540063  7.2721541  5.5493053 13.0319761
 [37]  5.4330042  2.9324986  3.4257384  2.0019504  5.9735743  4.5158558
 [43]  3.2299706 12.0112620  5.2986187  7.8859570  1.9491990  8.0865075
 [49]  4.0616294  7.3187900  8.1154061  6.1685047  8.1648742  5.2563961
 [55]  4.5739664  3.6205053  7.4141784  5.8869508  5.9728517  1.4717355
 [61]  1.1261881  1.9666790  8.9656340  3.8769728  1.8428999 11.0018934
```

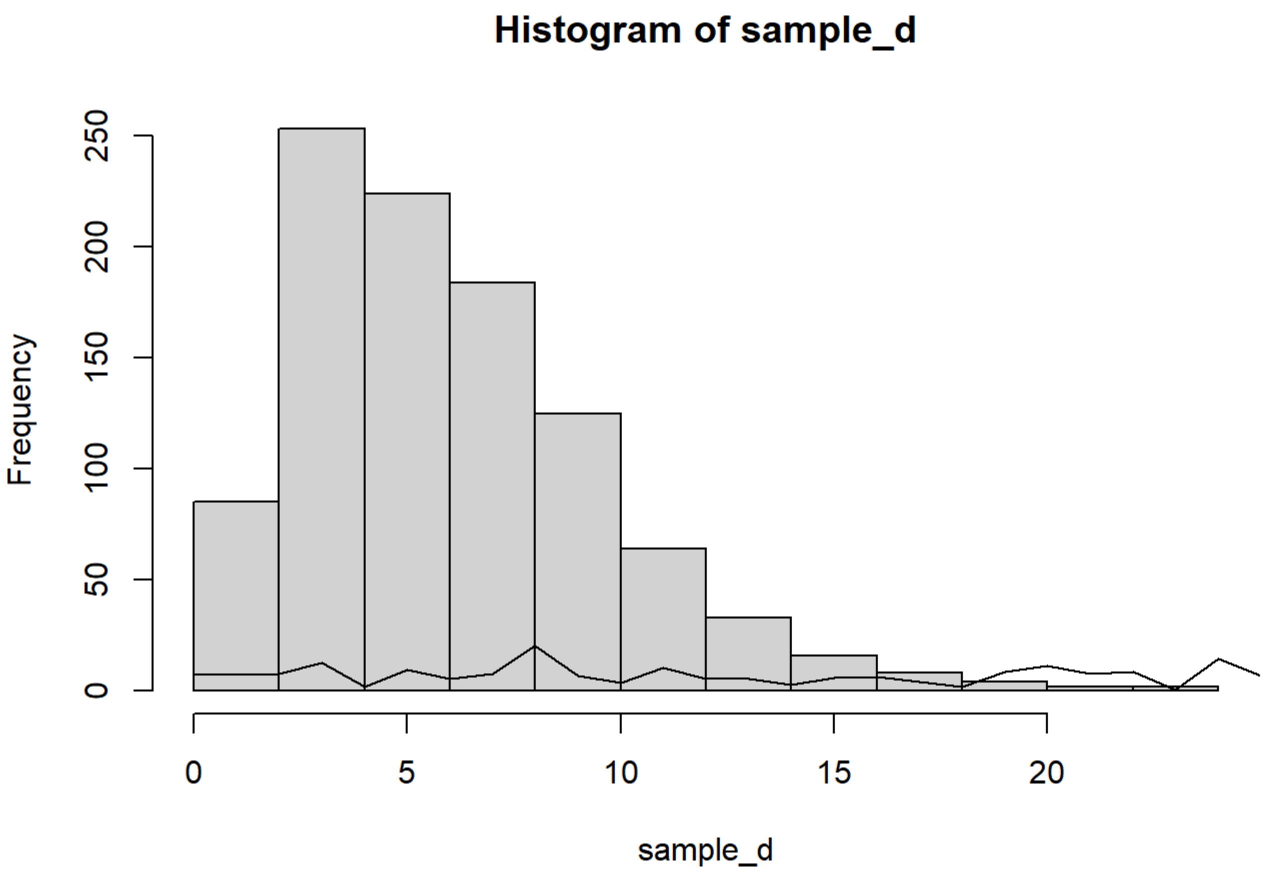
```
 [67] 10.4474478  3.5746800  6.2230784  5.8247693 10.5686527  2.2714452
 [73]  4.5439161  8.4389925  5.2867098  6.2992579  3.2916949 10.0226189
 [79] 12.9335990  3.3582087  1.8235704  8.2995843  5.7940020  3.1971039
 [85]  7.7188103  4.2462199  2.4821121 12.0136226  4.1950296  6.8511412
 [91]  2.2620584  8.8850650  6.1569979  6.7078523  4.3916118  5.4005168
 [97]  9.8551437  2.8586053  7.9950177 10.2432164  1.1354991  6.3954378
[103]  4.1862154 10.5618535  8.9448215  4.6105463  4.6052695  2.7502141
[109]  4.1998661  4.0806032  7.1065070  7.0349673  1.5434924  6.1543413
[115]  2.9881088  2.9209376  6.3654917  5.8144985 11.3137509 15.8173691
[121]  5.8372958  8.0748613  4.4954234  2.0337769  1.4895931  5.5756912
[127] 12.7462426  4.6060035 11.6307082  6.4886332 13.2758161  2.5080712
[133]  7.9750325  3.3110303  4.4776226  4.1139603  9.6998768  6.9500746
[139]  6.7068708 10.4002911  5.1923102  6.4757428  6.9818208  2.8664761
[145]  3.3504439  9.3562162  3.0164185 12.8431932  9.4124524  5.7712743
[151]  5.0013108  3.7033116  2.4034683  2.6339781  6.5810331  1.6598949
[157]  6.1961225  2.5745885  3.8895546 13.0535451  5.3104619  8.2026829
[163]  6.2953580  7.7953171 12.4506819  6.6470611  3.0980908  5.2177523
[169]  5.7323254  8.2932121  3.2976836  3.9053687 16.5575940 11.4658489
[175]  7.5453414  5.4941016  8.7154325  5.1838460  5.4566798  7.5918860
[181]  3.3498810  6.0283607  3.8796476  6.7846696  5.4560131 10.9988998
[187]  1.6322509  4.9366144  7.6527422  2.6078887  5.8955702  3.2909346
[193]  4.4521435  6.0135356  2.7996574  2.4763490  7.6349312  5.1720040
[199]  5.9764925  4.5043281  3.4846969  5.0396622 12.6317724  4.3206833
[205]  4.5165836  5.7167740  5.5369047  3.1607577  3.8866738  8.1577840
[211]  3.4332379  5.2179969  2.7019046 13.4160000  7.9180428  8.0808276
[217]  7.7839209 13.2162509  7.1940163  7.8343932  8.1698692  8.2149243
[223]  3.2112052  4.7565523  7.3763646 12.2906770  1.6104576  3.8912058
[229]  7.1553779  7.2471008  7.4245065  8.0556267  7.7656743  1.8976464
[235]  9.7361108  4.8147700  4.1212725 16.2788038  5.0352751  2.5459039
[241]  3.5887604  5.8990460  8.4994645  8.9404684  2.1995463 14.8317402
[247] 11.7935047  3.6237116  2.9446702  9.2502862  2.3641636  1.2742287
[253]  4.9511711  5.1783294  3.7100025  4.7099703  9.5460757  5.4894520
[259]  6.0277245  9.7658177  5.3457688  5.3405735  3.4533854  4.9747977
[265]  8.2855320 14.6534642  3.0805568  2.1043685  8.6823499  3.9368346
[271]  5.9304346  9.1901440  2.5760024  4.0572424  8.7025533 11.8736270
[277]  6.7072206  0.6675131  3.8372040  2.5142150  1.6215339  2.5552712
[283]  3.9363041  6.5694994  4.7930383 12.0486425  3.9936393  2.0073344
[289]  2.3641576  8.7306228  8.6229639  4.6635350  4.6494247  4.7508670
[295] 13.7592425  4.2871859  4.3471671  2.4648516  4.3018489  3.7775599
[301]  4.2955583  8.2097480  4.6013691  9.2045452 10.4071920  3.2487630
[307]  5.7527684  1.9179120  6.6374750  2.8966236  2.7631509  9.2877669
[313]  7.5103453  3.6184797  0.6160995 16.7558805  4.4741471  8.0067495
[319]  9.0282179  5.5609614  4.7189651  5.4844153  8.7011945  7.8649032
[325]  5.3224983 13.9729230  3.1523594 11.7349783  1.3552876  9.9745062
[331]  3.8993346  1.6312636  3.9367345  5.0102431  3.1256310  2.8113144
[337]  7.4013521  1.5273402  9.5366319  2.2926969  7.8915240  3.6031750
[343] 11.0321533  4.1249567 11.8458259  4.6056296  5.8286644 13.2193241
[349]  4.6976827  9.4070441  3.1799539  2.8467202  3.7904718  6.0667947
[355] 10.5400180  4.5729016  5.7728877  1.3691660  0.4234088  3.0038414
[361]  1.3963840  9.7057241  4.5508568  4.5549777  3.1855273  9.4712207
[367]  3.9679498  9.5869731  1.9421838 10.7955417  6.0084757  4.9548748
```

```
[373] 11.5161895  6.3920374  8.0656713  2.7937346  1.4951911  5.9467668
[379]  1.8760536  2.6595317  1.7400389  1.9662104  8.6671416  7.1220561
[385]  1.1979878  4.7257349  7.4507863  3.8323048  6.4737732  2.7971274
[391]  5.1356963 14.1441711  4.5527913  4.2540567  3.0642122  7.9401803
[397]  2.3393912  3.6561460  5.1125774  8.4556930  7.1342285  7.9563499
[403]  3.7499803  4.0866906  1.7996957 13.6149631  2.7088334  4.5671970
[409]  1.9063536  8.5782527  5.9406169  6.0643510  4.2808106  5.0451027
[415]  2.2885374  5.0851156  2.7851103  4.2009189  3.6319495  7.3934405
[421]  3.7255824  4.2584499  2.6937071 10.3619950  9.0018401  3.1043421
[427] 10.4808420  7.3685507 12.3344140  8.3659292  8.4450835  4.7049801
[433]  3.2167507  4.4508204  2.8829972  6.7538769  6.2643076  2.3509792
[439]  2.1941002  2.1097291  4.2185099 10.1170729 11.4634264  2.8174579
[445]  7.3198263  3.4651832  2.6841449  1.8326135  5.4547981  3.6509690
[451]  5.3998626  4.9646140  5.9855769  8.3866943  9.2503974 11.4282775
[457]  9.5389834  2.3037997 18.2442424  8.5601736  6.1389342  8.5710045
[463]  6.1287632  2.5936210  7.5501896  5.0847183  8.3522743  8.8948446
[469]  0.5128094  2.7172149  3.5187754  2.9669661 16.3715507  9.2946260
[475]  3.8090213  2.5813112  2.0593381  4.8509428 10.8959059 17.0593883
[481]  3.1901167  9.4403763  4.8786245  3.6820311  3.8590163 10.9749617
[487]  7.5576346  1.7949132  5.2004646 11.3155680  6.3026377  4.9308400
[493] 14.2058029  5.8619599  1.1827587  9.2016334  5.4584599 20.7070784
[499]  3.3794320 13.2065427  2.4570249  3.0827063 11.4122688  9.4816680
[505]  6.2537926  7.4891553  9.9364708  7.3372262  0.7256331  4.1805588
[511]  8.1668505  3.6886867  1.9355877  3.3375100  3.5055706  2.5622719
[517] 10.2443259  0.9598186  6.3002481  6.3788040  2.6076492  5.0673494
[523]  2.7072547 13.4309262  9.6173547  3.2032339  3.9429956  4.4970381
[529]  2.1874792 11.5635727  5.1395573  7.5928150  9.2275993  3.6802796
[535]  3.3870987  7.4417553  8.3325275  7.1525654 11.2378996  5.7630976
[541]  5.0527412  9.1067372 13.5347707  6.1292126  6.8166395  7.8022235
[547]  9.0757017 16.0993409  1.9780738  3.1862844  4.8737582  8.0236250
[553]  6.9502632  4.9305102  7.3715617  6.5356563  3.1188383  9.9645961
[559]  6.8600218  3.5952294 12.1812186  5.2921550  6.1715372  4.9226509
[565]  9.5786203  3.0565899  2.1257538  7.7777626  6.6411831  3.8179068
[571]  7.4578196  7.1512057  8.3445680  3.0322548  8.7600631  4.9393893
[577] 23.1243469  8.8639733  3.5354120  5.7534234  1.9030139  2.2585646
[583]  5.9235005  7.2389846  6.3530310  4.5050090  1.8585112  7.3375437
[589]  9.8541140  2.1803857  2.9428924  6.5120534  9.2367484  5.0353769
[595]  6.6894206  7.0985104  5.9739953  3.1037352  3.1802603  4.8551898
[601]  2.6465933  5.7473910 11.8696658  6.0445672 11.3419919  6.4202485
[607]  6.3509282 11.8154419  3.2570132 12.1489601  3.5329846  9.3991797
[613]  7.6284875  1.1779210  0.9948764  4.9302067  6.4265145 14.7549787
[619]  6.1305597  6.3211815  3.4992077 10.5381312  5.3025773  2.8795838
[625] 12.3005573  8.7068440  7.2460416  2.4759468  5.0280111  6.1299702
[631]  5.6386490  7.3671484  6.3755851  8.5957383  7.3647741  2.8917935
[637]  4.1844522  9.6896588  2.2736637  2.3611561  0.8051982 10.6912121
[643]  4.3780449  6.4544248  0.9789661 10.4620298  1.3708785  2.4015243
[649]  6.9187596  3.5386569  5.8938715 14.2982825  0.5260346  8.2778345
[655]  7.0607935  5.0278756  2.9422570  8.4991523  7.3422640  5.7860575
[661]  1.1699250  6.0116529  6.5278950  5.6762793  3.9194682  2.3855495
[667]  1.4843081 11.8625557  6.9490462  4.4700297  3.8445227 11.4460772
[673]  6.7059988  7.7426674  6.1367760  8.2380990  9.1374548  7.3593483
```

```
[679]  3.3536745  1.1869084  8.6740751  5.3324338  7.3195028  2.8649482
[685]  5.5095683  5.2853478  5.0974431  6.6831135  1.2248192  5.6197610
[691]  2.1131360  9.7241686  1.2440213 12.8676724  2.3198576 13.3046832
[697]  8.8721337  4.6032896  0.8413599 14.2270023  2.4688009 10.2167907
[703] 11.0133728  6.3030357  7.2771107  7.4251980  3.8915081  4.2686058
[709]  6.2642690  8.2890499  4.6763139  1.9270346  3.3553901  7.9515852
[715]  2.7584963  3.2122382  4.0936778  6.4343999  4.9575405 15.4141721
[721]  5.1463654  8.2073908  6.2919215  7.9279478 10.7279691  8.3246826
[727]  2.4897996  3.3139951  2.1691412 10.1521584  6.1032955  8.5735155
[733]  2.3699778  2.9716733 13.7887796  6.7289453  3.2452806 10.2830338
[739]  5.5814649  2.0764658 15.6569470  3.0517660  6.2359043  1.5733085
[745]  5.8320937  7.6323473  5.3461676  1.9599502  6.0388818 10.6903014
[751]  4.5822219  1.1745549 10.6030608  1.6751831  3.3105025 10.7331794
[757]  3.2027841  5.7302379  4.7112712  3.4186515  1.4522592  6.2165362
[763]  4.3145153  6.6333208  3.2654294 19.4486876  5.3792808  5.5771399
[769]  1.8465489  3.8371307  5.1012561  7.7532080  6.4062774  8.1465992
[775]  9.5777642 10.1258691  5.9121533  4.3464356  4.3614588  6.4175546
[781]  2.1068070  3.2581899  2.6450222  9.4239262 15.9704372  2.6858201
[787] 10.7407495  6.2442384  6.7990372  7.6091630  3.7595180 12.9787665
[793]  1.6922229  5.6301523  6.0107079  9.9800781  4.2238547  4.2202432
[799]  2.6174015  8.3060782  8.2481008 13.5619392  3.2260177 17.8370371
[805] 10.1502495 10.8683191  3.1028714  2.7765468  3.3839155  4.1759743
[811]  8.9618833  3.0149416  3.0945488  9.1082219  4.3458205  5.4374527
[817]  5.8279077 18.9359847  5.0205383  1.9312148  7.7702759  1.3245852
[823]  7.2922465  9.1338655  9.7340053  4.9267447  6.1246842  2.4357077
[829]  4.5999734  1.1966494 13.5727629  3.2125511  4.8176579  4.1004453
[835]  6.9887293  3.3153844  3.6552596  1.9280199 10.7334057  1.9882751
[841]  2.7404322  8.6428120  6.4071802  6.3814346  4.4257295  8.6818610
[847] 15.0711859  4.9607229  1.7716452  3.4373777  5.0955921  3.4081607
[853]  3.9830022  7.6651822  3.9351972  1.3830661 11.6472355  3.2520711
[859]  2.9556407  2.1642431  0.7967141 13.8629624  7.7218302  9.9213554
[865]  4.4729268  3.8389781  4.1968386  2.2990535  5.6034324  9.6451014
[871]  2.8573212  5.6743685  2.1045521  4.8035946  8.1342117  2.3705498
[877]  3.0858569  7.2621644  6.8705961  6.0521742  8.0355099  4.5056412
[883]  4.2918884  3.0984919  5.8262319  6.4497499  3.5577309  8.4591712
[889]  2.3905314  6.7172453 18.7236187  9.1683034 10.4117690  2.1896512
[895]  4.1376697  4.1031296  3.0478956  5.4897983  8.3017385  1.8573540
[901]  3.3551325  9.9686164  2.6039786  6.9484387  3.9014241  3.1205543
[907]  3.8809317  4.6330312 10.9305822  4.3681421  4.8860766  6.6504452
[913]  4.7210977 10.3264834  2.0574045  1.2444220  9.3675041  8.7350248
[919]  1.3877980  3.6531904  7.3826854  4.8962689  4.3773035  4.4102098
[925]  9.9397336  7.3423912  7.9642122  3.6598193 11.1508928  2.7652808
[931]  5.9457058  4.0075096  9.5160638  7.7874803 13.6402753 11.4769703
[937]  3.6081657  7.6973610  3.5356194  6.1907692 15.7004412  1.9674330
[943]  8.4337901  5.7593042  6.2871552  3.7305215  3.3358729  3.2154215
[949] 15.5350328  6.7263765 14.2946931  0.7913834  5.4141601  4.4475887
[955]  1.8882244  8.4493825  9.3804287  7.4688734  3.7437141  3.6058504
[961]  5.7710576  6.4450959 23.1208683  7.4958413  3.7712302  7.5068970
[967]  6.1645247  2.3377568  2.3476464  2.7536226 15.0645973  7.6316485
[973]  2.5559401  6.7497755 10.5277922  9.5311215  1.8372693  8.4885206
[979]  5.1239763  2.4141813  1.9305211  3.9176818  3.3341995  1.0332564
```

```
[985]  1.3875498  6.6126469  3.5107163  2.2667876  9.6624123  3.0833496
[991]  1.7512639 11.4677746  4.9944483  4.7900105  2.4384557  5.3233732
[997]  6.0811645  6.0652547  3.3406984  5.6534930
```

```r
hist(sample_d)

lines(0:999, rgamma(1000, shape = 3, scale = 2))
```



Arsenic

a

```r
# Expectation in statistics means "mean"
# In the normal distribution the mean = median = mode



mean <- 5
s_d <- 1

# X ~ N(mu, sigma^2)

```

```
# SE = sigma^2/n

# X ~ N(5, 1/4)

# X ~ N(5, 0.25)
```

b

```
pnorm(7, mean = 5, sd = 1,
      lower.tail = FALSE)
```

```
[1] 0.02275013
```

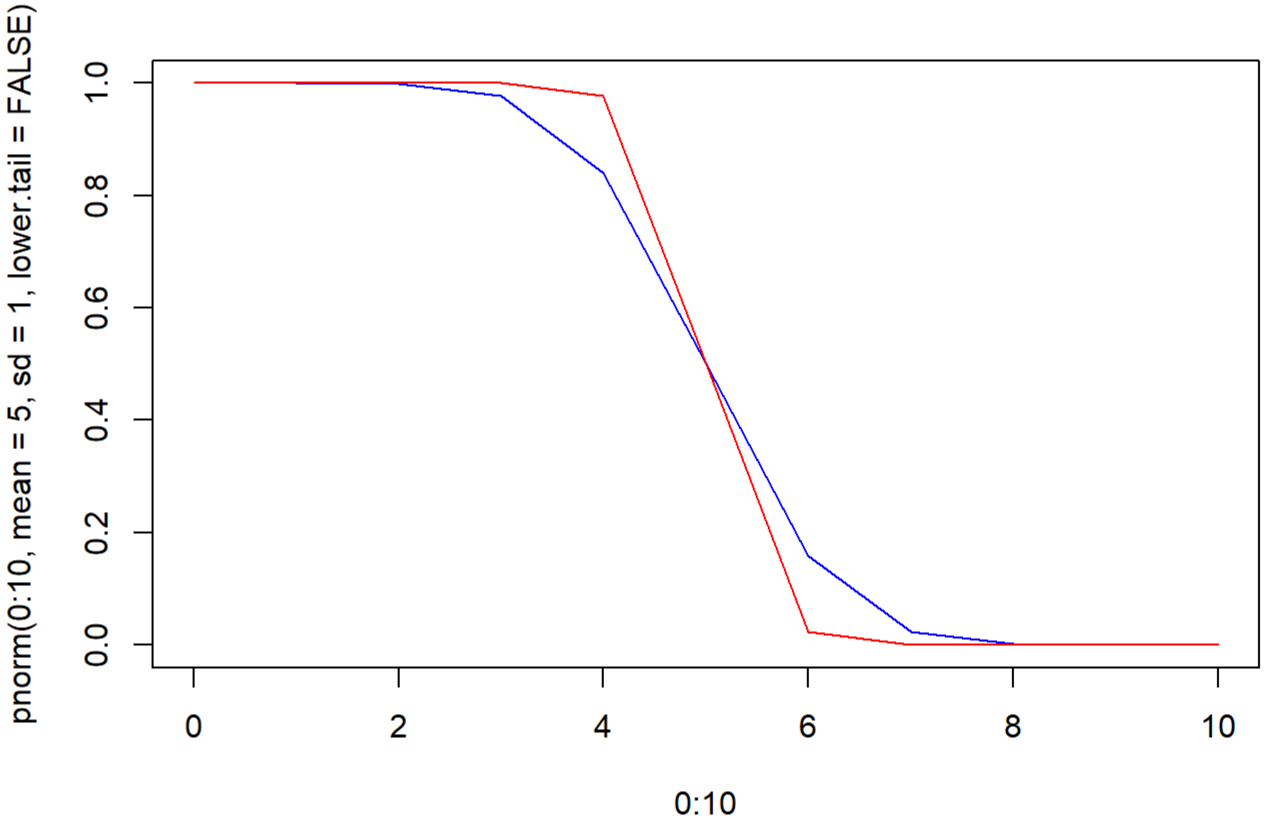
c

```
pnorm(7, mean = 5, sd = sqrt(0.25),
      lower.tail = FALSE)
```

```
[1] 3.167124e-05
```

d

```
plot(0:10, pnorm(0:10, mean = 5, sd = 1,
                 lower.tail = FALSE),
     col = "blue", type = "l")

lines(0:10, pnorm(0:10, mean = 5, sd = 0.5,
                  lower.tail = FALSE),
      col = "red")
```



Groundwater

```
mean <- 20
s_d <- 5

pnorm(30, mean = 20, sd = 1, lower.tail = FALSE)
```

```
[1] 7.619853e-24
```

Regulation

a

```
## Regulation

n <- 365
p <- 1/365

# For binomial distribution: Mean = np, SD = np(1-p)
```

```r
expectation <- n*p
expectation
```

```
[1] 1
```

b

```r
# Yes the distribution of emissions satisfy the regulation?
```

c

```r
x <- 2
n <- 365
p <- 1/365

pbinom(1, size = n, prob = p, lower.tail = F)
```

```
[1] 0.2642409
```

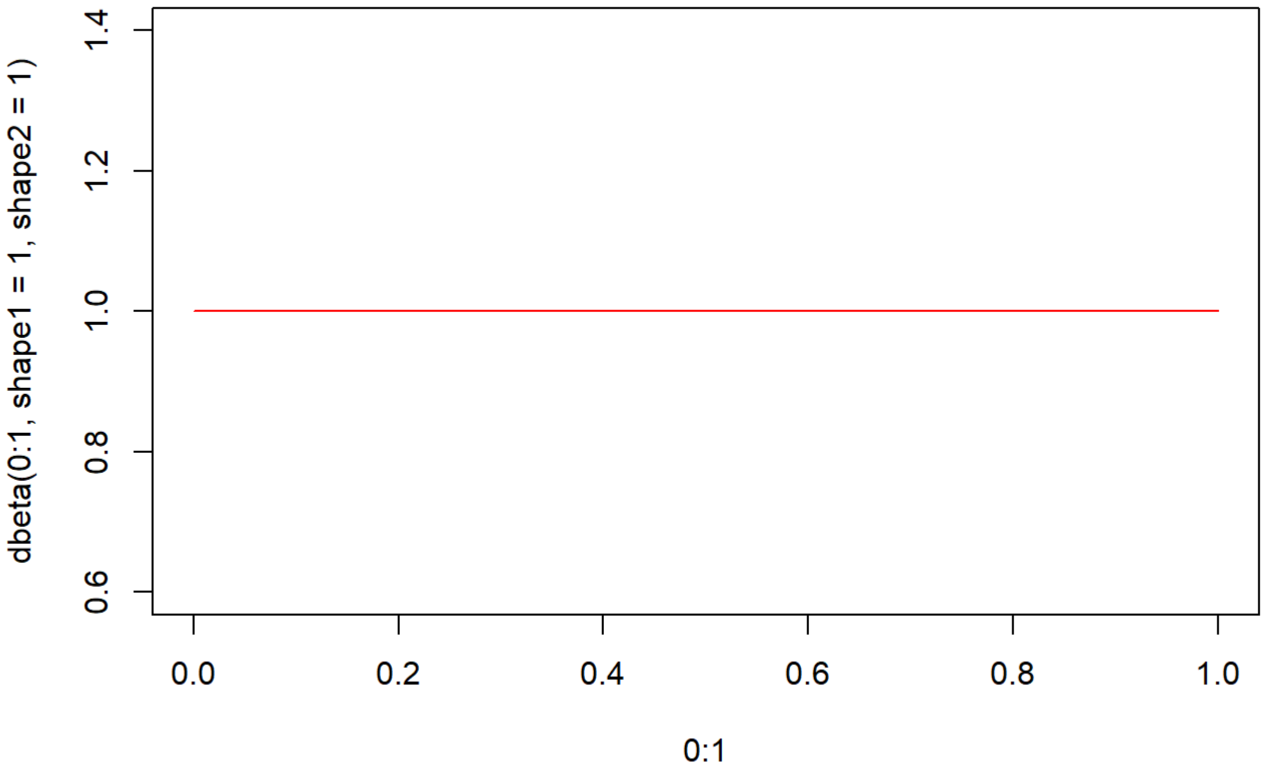
Poisson Distribution

```r
# The only parameter of a poisson distribution "lambda"
# lambda = np

lambda <- 1


ppois(1, lambda = 1, lower.tail = FALSE)
```

```
[1] 0.2642411
```

Uniform and Beta Distribution

```r
plot(0:1, dbeta(0:1,shape1 = 1, shape2 = 1),
     type = "l", col = "blue")


lines(0:1, dunif(0:1), col = "red")
```



Exponential Distribution

```
# a
# P(Y = 0.5) = 0

#b
pexp(1, rate = 2, lower.tail = FALSE)
```

```
[1] 0.1353353
```

```
#c

# Undefined because -1 is less than our range which is [0,infiniti]

#d

# Undefined because 0.2 is less than 0.7

#c
pexp(0.6, rate = 2) - pexp(0.1, rate = 2)
```

```
[1] 0.5175365
```

Species Dispersal

```r
dgeom(2, prob = 0.17)
```

```
[1] 0.117113
```

Glaucouse winged gull

```r
dnbinom(1, size = 3, prob = 0.76)
```

```
[1] 0.3160627
```